

Data Structures Using C

Lecture 1

Introduction to Data Structures, ADTs and Applications

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What is Data?

Definition

Data is a collection of facts, numbers, characters or values that can be processed to generate meaningful information.

Examples:

- Student Marks
- Employee Salary
- Product Price
- Mobile Number
- Temperature Reading

Information = Processed Data

What is a Data Structure?

Definition

A Data Structure is a systematic way of organizing, storing and managing data so that it can be accessed and modified efficiently.

Goal

- Faster Searching
- Efficient Insertion
- Efficient Deletion
- Better Memory Utilization
- Improved Performance

Why Do We Need Data Structures?

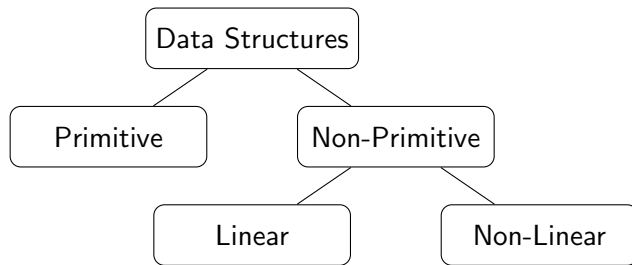
Without proper data structures:

- Programs become slow.
- Memory usage increases.
- Searching large data becomes inefficient.
- Difficult to maintain applications.

Real-life Example

- Phone Contacts
- Google Maps
- Instagram Feed
- Banking Systems
- Hospital Management

Classification



Primitive:

- int
- char
- float
- double

Non-Primitive:

- Arrays
- Linked Lists
- Stacks
- Queues
- Trees
- Graphs

Linear vs Non-Linear Structures

Linear

- Sequential arrangement
- Single level
- Easy traversal

Examples:

- Array
- Stack
- Queue
- Linked List

Non-Linear

- Hierarchical
- Multiple relationships
- Complex traversal

Examples:

- Trees
- Graphs

Operations on Data Structures

Every Data Structure supports several operations.

Operation	Purpose
Traversal	Visit all elements
Insertion	Add element
Deletion	Remove element
Searching	Find element
Sorting	Arrange elements
Merging	Combine structures

Abstract Data Type (ADT)

Definition

An Abstract Data Type (ADT) defines:

- What operations can be performed
- Not how they are implemented

Examples:

- Stack
- Queue
- List
- Priority Queue

Implementation can be done using:

- Arrays
- Linked Lists

ADT Example: Stack

Operations:

- push()
- pop()
- peek()
- isEmpty()
- isFull()

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Applications of Data Structures

Data Structure	Applications
Array	Matrices, Images
Linked List	Music Playlist
Stack	Undo, Browser History
Queue	Printer Queue
Tree	File System
Graph	Social Networks, Maps
Hash Table	Dictionaries

Good Algorithm + Good Data Structure = Efficient Program

Choosing an inappropriate data structure may increase

- Execution Time
- Memory Usage
- Program Complexity

Can an Array grow dynamically?

If not,

How does a Linked List solve this problem?

We'll discuss this in the next lecture.

Questions?

Thank You